Generic backward uniqueness for linear retarded equations

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Abstract

We prove residual backward uniqueness for all pairs of real times in the specified compact C1 coefficient topology. A continuous evolution on a Hilbert phase space makes injectivity a countable intersection of open conditions, while a dense gap family supplies the one-to-one solution maps.

Repository target. This manuscript concerns AIM problem 921 at revision `5d7e26e8444e`. The precise scope is stated below.

1 Main result and proof

Answer. The set in Problem 921 is residual. The useful change of phase space is to retain the current value separately and put the past history in L^2. The C^1 dependence of the history functional in operator norm gives a common finite measure dominating all delay measures on a compact time interval. This makes the enlarged Hilbert-space evolution well defined and continuous in the coefficient. Injectivity is a G_δ condition for operators on a separable Hilbert space. The gap truncations behind the density theorem of Hale–Oliva [1] are injective on this enlarged space as well.

Fix $r > 0$ and an integer $n \geq 1$, and use exactly the coefficient space and topology in the question:

$$C_r = C([-r, 0]; \mathbb{R}^n), \qquad \mathcal{B}_r = \mathcal{L}(C_r, \mathbb{R}^n), \qquad \mathcal{X} = C^1(\mathbb{R}; \mathcal{B}_r).$$

Introduce the separable real Hilbert space

$$\mathcal{E} = \mathbb{R}^n \oplus L^2([-r, 0]; \mathbb{R}^n). \tag{921.1}$$

The two components of an initial state (a, ϕ) are independent: no point value is assigned to the L^2 class ϕ.

Theorem 1.1. *There is a countable intersection $\mathcal{R}$ of open dense subsets of $\mathcal{X}$ such that, for every $L \in \mathcal{R}$, its enlarged evolution $U_L(t, \sigma)$ on $\mathcal{E}$ is injective for every real $\sigma < t$. Consequently the set $\mathcal{G}_{r,n}$ in Problem 921 is residual in $\mathcal{X}$.*

1.1 A common dominating measure

By the Riesz representation theorem, write

$$L(t)f = \int_{[-r,0]} d\mu_t(\theta) f(\theta),$$

where $\mu_t = (\mu_{t,ij})_{i,j=1}^n$ is a matrix of finite real signed measures. The norm $\|\mu\|_{\mathrm{var}} = \sum_{i,j} |\mu_{ij}|([-r,0])$ is equivalent, with constants depending only on n, to the operator norm of L. Thus $t \mapsto \mu_t$ is C^1 in variation norm.

Fix a compact time interval $I = [a,b]$. Define the finite positive measure

$$\nu = \sum_{i,j} \left(|\mu_{a,ij}| + \int_a^b |\mu'_{s,ij}|\, ds \right). \tag{921.2}$$

The integral is an integral of finite measures in variation norm; the variation map is continuous in that norm. The fundamental theorem of calculus and the triangle inequality for variation give

$$\sum_{i,j} |\mu_{t,ij}| \le \nu \qquad (t \in I). \tag{921.3}$$

In particular, the matrix densities $R(t,\theta) = d\mu_t/d\nu$ may be chosen jointly measurable, and

$$\|R(t,\theta)\|_{\mathrm{op}} \le \sum_{i,j} |R_{ij}(t,\theta)| \le 1 \quad \text{for almost every } (t,\theta) \text{ with respect to } dt\, d\nu(\theta).$$

For justification of joint measurability, the densities form a continuous map from I to $L^1(\nu)$ entry by entry, because the corresponding measures are continuous in variation norm. A jointly measurable representative of a strongly measurable L^1-valued function suffices.

The same construction applies to a difference of two coefficient families. On $[a,b]$, its envelope ν_Δ satisfies

$$\nu_\Delta([-r,0]) \le \|\mu_a - \widetilde{\mu}_a\|_{\mathrm{var}} + (b-a) \sup_{s\in[a,b]} \|\mu'_s - \widetilde{\mu}'_s\|_{\mathrm{var}}. \tag{921.4}$$

This is controlled by a constant times the compact C^1 distance in the problem statement.

1.2 Evolution on the Hilbert phase space

Fix an initial time σ and a terminal bound $b > \sigma$. Given $(a,\phi) \in \mathcal{E}$, extend ϕ by zero outside $[-r,0]$ and define the initial-history forcing, for almost every $s \in [\sigma,b]$, by

$$F_{L,\phi}(s) = \int_{[-r,0]} d\mu_s(\theta)\, \mathbf{1}_{\{s+\theta<\sigma\}} \phi(s+\theta-\sigma). \tag{921.5}$$

Using an envelope from (921.2) on $[\sigma,b]$, Minkowski's inequality and translation invariance of Lebesgue measure give

$$\begin{aligned} \|F_{L,\phi}\|_{L^2([\sigma,b])} &\le \int_{[-r,0]} \left(\int_\sigma^b \mathbf{1}_{\{s+\theta<\sigma\}} |\phi(s+\theta-\sigma)|^2\, ds \right)^{1/2} d\nu(\theta) \\ &\le \nu([-r,0])\, \|\phi\|_2. \end{aligned} \tag{921.6}$$

This also proves independence of the representative of ϕ. Indeed, translating a Lebesgue null set produces a null set for each fixed θ, and Fubini's theorem applies to $ds\, d\nu(\theta)$.

For a continuous future path $x : [\sigma,b] \to \mathbb{R}^n$, put

$$(B_L x)(s) = \int_{[-r,0]} d\mu_s(\theta)\, \mathbf{1}_{\{s+\theta\ge\sigma\}} x(s+\theta).$$

This is measurable and depends only on values of x up to time s. If $M = \sup_{s\in[\sigma,b]} \|\mu_s\|_{\mathrm{var}}$, then

$$|(B_L x)(s)| \le M \sup_{\sigma\le u\le s} |x(u)|. \tag{921.7}$$

The Volterra integral equation

$$x(t) = a + \int_\sigma^t F_{L,\phi}(s)\,ds + \int_\sigma^t (B_L x)(s)\,ds \tag{921.8}$$

therefore has a unique continuous solution, by Picard iteration on short intervals or the usual factorial estimate for Volterra iteration. It belongs to $W^{1,2}([\sigma,b];\mathbb{R}^n)$ and obeys, with $T = b-\sigma$,

$$\sup_{\sigma\le t\le b} |x(t)| \le (|a| + \sqrt{T}\,\nu([-r,0])\|\phi\|_2)e^{MT}. \tag{921.9}$$

Concatenate the initial L^2 history and this future path, and define

$$U_L(t,\sigma)(a,\phi) = (x(t), x_t), \qquad x_t(\theta) = x(t+\theta) \quad \text{as an } L^2 \text{ class.} \tag{921.10}$$

Equations (921.6)–(921.9) make this a bounded linear operator on $\mathcal{E}$. A change of the concatenated path at a single joining point has no effect on the equation almost everywhere: this follows from the same Fubini argument used for representative independence. Uniqueness then gives the evolution law

$$U_L(t,s)U_L(s,\sigma) = U_L(t,\sigma) \qquad (\sigma \le s \le t). \tag{921.11}$$

For each initial state, $U_L(t,\sigma)(a,\phi)$ is strongly continuous in $t \ge \sigma$, including at $t = \sigma$. At the initial time this uses $x(t) \to a$, strong continuity of translations in L^2, and the fact that the new future segment has length $t - \sigma \to 0$ and bounded values.

Lemma 1.2. *For each fixed pair $\sigma < t$, the map $L \mapsto U_L(t,\sigma)$ from $\mathcal{X}$ into $\mathcal{L}(\mathcal{E})$ is continuous in operator norm.*

Proof. Apply (921.6) to the difference of two coefficient families:

$$\|F_{L,\phi} - F_{\widetilde{L},\phi}\|_2 \le \nu_\Delta([-r,0])\|\phi\|_2.$$

For solutions with the same initial state, their difference $z = x - \widetilde{x}$ satisfies the difference of (921.8), with

$$B_L x - B_{\widetilde{L}}\widetilde{x} = B_L z + (B_L - B_{\widetilde{L}})\widetilde{x}.$$

The last term has absolute value at most $\nu_\Delta([-r,0])\sup|\widetilde{x}|$. Use (921.9) for $\widetilde{x}$ and Gronwall's inequality for z. On a neighborhood of L with a uniform bound in $C^1([\sigma,t];\mathcal{B}_r)$, this gives

$$\sup_{\sigma\le s\le t} |z(s)| \le C\,\nu_\Delta([-r,0])(|a| + \|\phi\|_2),$$

where C is independent of the initial state and the nearby coefficient. The difference of the terminal histories is at most $\sqrt{r}\sup|z|$ in L^2, since their older initial-history parts agree. Now use (921.4) and take the supremum over the unit ball of $\mathcal{E}$. □

1.3 A dense family injective on the enlarged space

Consider coefficients of the form

$$L(t)f = A(t)f(-r) + \int_{[-r+\rho,0]} d\eta_t(\theta) f(\theta), \qquad 0 < \rho < r, \tag{921.12}$$

where $A(t)$ is invertible for almost every real t and the coefficient family is C^1 in variation norm.

Lemma 1.3. *Every family (921.12) has injective $U_L(t,\sigma)$ on $\mathcal{E}$ for every $\sigma < t$.*

Proof. Suppose $U_L(t,\sigma)(a,\phi) = 0$. Forward uniqueness extends the future path by zero after t, so the concatenated path is zero almost everywhere on $[t-r,\infty)$.

We show how to extend a zero region backward. Suppose that the path is zero almost everywhere on $[b,\infty)$, with $b \geq \sigma - r$. For almost every

$$s \geq \max\{\sigma, b + r - \rho\},$$

its derivative is zero, and the integral in (921.12) is zero because all arguments $s+\theta \geq s-r+\rho \geq b$. The assertion about the integral is an almost-everywhere assertion justified by a common dominating measure and Fubini, exactly as in (921.6). The equation gives $A(s)x(s-r) = 0$. Since $A(s)$ is invertible almost everywhere, the path is zero almost everywhere on

$$[\max\{\sigma - r, b - \rho\}, \infty).$$

Starting with $b = t - r$, finitely many iterations give zero on $[\sigma - r, \infty)$. Thus $\phi = 0$ in L^2. The future path is continuous at σ, so its almost-everywhere vanishing also gives $a = x(\sigma) = 0$. □

Lemma 1.4. *The families (921.12) are dense in $\mathcal{X}$.*

Proof. Start with any coefficient L, represented by μ_t, and any compact time interval on which it is to be approximated. Write $A(t) = \mu_t(\{-r\})$. For $\rho \downarrow 0$, restriction to $(-r, -r+\rho)$ tends strongly to zero on the space of finite measures in variation norm. The restriction operators have norm at most one. Their convergence is therefore uniform on each norm-compact set of measures. Apply this to the compact sets of values of μ_t and μ'_t on the chosen time interval. It follows that

$$\mu_t^\rho = A(t)\delta_{-r} + \mu_t|_{[-r+\rho,0]}$$

approximates μ_t in compact C^1 norm as $\rho \downarrow 0$.

It remains to perturb A so that it is invertible almost everywhere. For each fixed t, the polynomial $S \mapsto \det(A(t) + S)$ on $M_n(\mathbb{R})$ is not identically zero. Its zero set has Lebesgue measure zero. Fubini's theorem on $[-m, m] \times \{S : \|S\| < \epsilon\}$, followed by a countable intersection over positive integers m, shows that for almost every arbitrarily small constant matrix S,

$$\det(A(t) + S) \neq 0 \quad \text{for almost every } t \in \mathbb{R}.$$

Replacing $A(t)$ by $A(t) + S$ changes the coefficient by $f \mapsto Sf(-r)$ and does not change its time derivative. Combining this perturbation with the gap truncation proves density in precisely the required compact C^1 topology. □

1.4 The Baire argument and the original continuous histories

On a separable Hilbert space, the injective bounded operators form a G_δ in operator norm. Here are the details needed in this setting. Choose a countable dense set $D \subset \mathcal{E}$ and an orthonormal basis (e_j). For positive integers j, m, let

$$\mathcal{O}_{j,m} = \bigcup_{v \in D} \{T \in \mathcal{L}(\mathcal{E}) : \|T^* v - e_j\| < 1/m\}. \tag{921.13}$$

Each set is open. Moreover,

$$\bigcap_{j,m} \mathcal{O}_{j,m} = \{T : \overline{\operatorname{Ran} T^*} = \mathcal{E}\} = \{T : \ker T = \{0\}\}. \tag{921.14}$$

The first equality uses density of D and continuity of T^*; the second is the Hilbert-space orthogonal-complement identity.

For every rational pair $\sigma < t$, take the preimages of the sets (921.13) under $L \mapsto U_L(t,\sigma)$. They are open by Lemma 1.2. Each contains the dense gap family of Lemmas 1.3 and 1.4, so each is dense. Their intersection over all j, m and all rational pairs $\sigma < t$ is a countable intersection $\mathcal{R}$ of open dense subsets of $\mathcal{X}$. The latter is a Fréchet space, hence a Baire space.

For $L \in \mathcal{R}$, injectivity holds for every real pair as well. Suppose $U_L(t,\sigma)v = 0$ with real $\sigma < t$. Choose a rational $p > t$. For every rational $q \in (\sigma, t)$, the evolution law gives

$$U_L(p,q)U_L(q,\sigma)v = U_L(p,t)U_L(t,\sigma)v = 0.$$

The first factor is injective because p, q are rational. Thus $U_L(q,\sigma)v = 0$. Let rational $q \downarrow \sigma$ and use strong right continuity to conclude that $v = 0$.

Finally, the bounded injective map

$$J : C_r \longrightarrow \mathcal{E}, \qquad J\phi = (\phi(0), \phi)$$

identifies a continuous history with its compatible enlarged state. The classical continuous-history solution satisfies (921.8), so uniqueness gives

$$U_L(t,\sigma)J\phi = J\big(T_L(t,\sigma)\phi\big).$$

If $T_L(t,\sigma)\phi = 0$, enlarged-space injectivity implies $J\phi = 0$, and hence $\phi = 0$. Therefore $\mathcal{R} \subseteq \mathcal{G}_{r,n}$, proving Theorem 1.1.

The proof uses the original C^1 coefficient topology throughout. It does not infer openness of continuous-history injectivity, and it does not replace distributed delays by finite step measures. The gap family is used to prove density; the G_δ argument takes place on the larger Hilbert phase space before returning to the specified history space.

References

[1] J. K. Hale and W. M. Oliva, *One-to-oneness for linear retarded functional differential equations*, J. Differential Equations **20** (1976), 28–36. doi:10.1016/0022-0396(76)90093-0.